

Research paper

Hybrid explainable deep learning framework for skin disease detection and future risk prediction using Sakaguchi function with CNN-transformer models

Pavani Maram^a, Parvathi Ramasubramanian^{a,b}, Pattabiraman Venkatasubbu^{a,*}

^a School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu, India

^b Centre for Advanced Data Science, Vellore Institute of Technology, Chennai, Tamil Nadu, India

ARTICLE INFO

Keywords:

Skin disease detection
Dermatoscopic images
Sakaguchi function
Preprocessing
CNN
Vision transformer
Explainable AI
Grad-CAM
GANs
Risk prediction

ABSTRACT

Skin disorders, which include a variety of bacterial, fungal, viral, and parasitic illnesses, represent a major global health concern. Using dermatoscopic pictures, this study suggests an intelligent and comprehensible deep learning-based method for automated skin disease detection and risk prediction. Hair removal, bilateral filtering, K-means clustering for artifact suppression, and histogram equalization to improve lesion clarity are all part of the preprocessing procedure. In addition to this, the Sakaguchi function, mathematically expressed as $f(x) = x \cdot \exp(-\alpha x^2)$ where α is a tunable contrast-control parameter, is introduced as an advanced enhancement technique. Unlike conventional methods such as CLAHE, it adaptively enhances local contrast while preserving edges and minimizing noise amplification, thereby improving lesion texture visibility and segmentation accuracy. Features were extracted using pretrained CNN models like ResNet50 and InceptionV3, followed by hybrid deep learning architectures such as CNN-BiLSTM, CNN-Capsule Networks, and CNN combined with Vision Transformers (ViT). With an accuracy of 96.10 %, precision of 96.40 %, recall of 95.80 %, and F1-score of 96.10 %, the CNN + ViT model combined with Sakaguchi-based preprocessing produced the best results. Clinical dependability was improved by using explainable AI approaches as Grad-CAM, Layer-wise Relevance Propagation (LRP), and SHAP to visualize and analyze model decisions. GANs were used to generate synthetic data that enhanced performance on minority classes in order to remedy class imbalance. The Overall, the CNN-ViT hybrid framework's incorporation of the mathematically defined Sakaguchi enhancement shows theoretical and practical improvements in generalization, interpretability, and robustness, making it appropriate for dermatological diagnosis and progression monitoring in the real world.

1. Introduction

Skin conditions are among the most common health issues in the world, impacting people of all ages and backgrounds. The World Health Organization (WHO) estimates that over 900 million people worldwide suffer from skin conditions each year, making up roughly 2 % of all diseases. These disorders include a wide variety of infections, such as those caused by bacteria, viruses, fungi, and parasites, as well as long-term, non-infectious illnesses like psoriasis, vitiligo, and eczema.

An estimated 20-30 % of dermatology diagnoses are incorrect, mostly because of overlapping visual patterns and restricted access to qualified specialists, particularly in low-resource settings. This situation demonstrates the increasing demand for intelligent diagnostic sys-

tems that are not only accurate but also scalable, dependable, and interpretable in a variety of clinical settings.

However, the nonlinear lighting and textural irregularities seen in dermatoscopic pictures are frequently not sufficiently addressed by traditional enhancement methods such as histogram equalization, CLAHE, or Retinex transformations.

Initially, conventional innovations such as bilateral filtering for noise reduction, K-means clustering for artifacts, histogram equalization for contrast correction, and morphological hair removal and suppression were used.

Building on this, we amplified diagnostically important lesion features by including the Sakaguchi function, a nonlinear mathematical transformation recognized for adaptive local contrast enhancement. It improves Fisher separability and feature robustness by dynamically

* Corresponding author.

E-mail address: pattabiraman.v@vit.ac.in (P. Venkatasubbu).

URL: <https://www.vit.ac.in> (P. Venkatasubbu)

<https://doi.org/10.1016/j.rineng.2025.108489>

Received 11 August 2025; Received in revised form 13 November 2025; Accepted 28 November 2025

Available online 4 December 2025

2590-1230/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

highlighting fine-grained intensity fluctuations inside lesion boundaries, in contrast to traditional modifications.

As far as we are aware, this is the first time the Sakaguchi function has been incorporated into a whole dermatological deep learning pipeline, linking interpretability and picture improvement.

Because of its capacity to carry out nonlinear intensity transformation, which is defined as follows, the Sakaguchi function—which was first used in geometric function theory to depict bounded analytic mappings—has lately been modified for medical image enhancement.

$$I'(x, y) = \mu_{ROI} + \gamma(x, y) \cdot (I(x, y) - \mu_{ROI})$$

$$\gamma(x, y) = \beta + \alpha \tanh\left(\frac{I(x, y) - \mu_{ROI}}{\sigma_{ROI} + \varepsilon}\right)$$

This function adjusts contrast enhancement locally according to the lesion region's intensity deviation. The Sakaguchi function performs context-aware enhancement, selectively highlighting edges and pigment variations without enhancing background noise, in contrast to CLAHE, which uses uniform histogram-based contrast stretching. It works especially well for dermatoscopic images with poor contrast or variable lighting because of its distinctive nonlinear behavior. [1]

Additionally, explainability is stressed as a therapeutic and regulatory requirement rather than an optional element. Recent initiatives within the Digital Information Security in Healthcare Act (DISHA) framework and the National Strategy for Artificial Intelligence (NITI Aayog, 2021) highlight the significance of accountability, traceability, and transparency in healthcare AI systems in India. In order to maintain patient safety and physician trust, these regulations support interpretable and ethically controlled AI models, especially in high-stakes fields like dermatological diagnostics. In order to improve diagnostic transparency and support doctors in clinical decision-making for skin lesion analysis, our study incorporates Explainable AI (XAI) techniques—Grad-CAM, Layer-wise Relevance Propagation (LRP), and SHAP—to produce localized visual rationales for each prediction.

To ensure comprehensive learning, we created a composite dataset from several publicly available sources, including Kaggle's Skin-Disease-Dataset, filtered non-melanoma parts of HAM10000, ISIC archives, the Monkeypox Skin Lesion Dataset, and the Skin Disease MNIST dataset. These datasets were deliberately selected to focus on non-cancerous skin conditions in order to guarantee the system's usability in real clinical screening of common and underrepresented illnesses.

For feature extraction and classification, we employed pretrained convolutional neural network (CNN) designs such as ResNet50 and InceptionV3. We also looked into more complex hybrid deep learning models, including CNN-BiLSTM for spatiotemporal pattern recognition, CNN-Capsule Networks for viewpoint-invariant analysis, and a CNN combined with Vision Transformers (ViT) for global attention.

In conclusion, by providing a mathematically sound, comprehensible, and broadly applicable paradigm appropriate for practical application, this study closes a significant gap between explainability and accuracy in dermatological AI.

2. Literature review

Skin disease detection has become more important in the medical field due to the global incidence of dermatological illnesses and the increasing requirement for accurate and timely diagnosis [2]. Numerous studies have investigated deep learning, traditional machine learning methods, and domain-specific enhancements to improve detection performance and interpretability.

Hybrid feature selection techniques like Chi-Square, Information Gain, and Principal Component Analysis (PCA) have demonstrated significant improvements in classifier performance in traditional machine learning approaches, especially when combined with ensemble techniques like Bagging, Boosting, and Stacking [3–5]. Additionally, the

classification process has produced excellent accuracy when physiologically relevant clinical variables, such as erythema, itching, and other dermatological markers, are incorporated; some studies have reported using Stochastic Gradient Descent (SGD) to achieve up to 99.09% accuracy [6].

Deep learning methods have also advanced significantly. Lightweight designs like MobileNetV2 in combination with Long Short-Term Memory (LSTM) networks have shown potential for deployment in low-computation scenarios with successful sequence modeling of illness development [7,8]. On test datasets, hybrid models that use DenseNet121 and EfficientNetB0 have shown outstanding generalization performance, reaching up to 97.57% accuracy [9,10]. In order to overcome class imbalance and overfitting, more sophisticated designs such as EfficientNetV2L have profited from data augmentation techniques, such as the creation of synthetic data using Generative Adversarial Networks (GANs) [11,12].

Preprocessing is essential to enhance image quality and facilitate more effective feature extraction. Convolution-based image enhancement utilizing Mittag-Leffler-type distributions and Geometric Function Theory (GFT) [1,13] are examples of mathematical enhancement techniques that have improved image contrast and detail. Low-light image enhancement metrics like PSNR and SSIM have significantly improved thanks to recent advancements employing Gegenbauer polynomials and Sakaguchi-type functions [14,15]. To improve lesion delineation and boundary identification, Sakaguchi mappings based on edges have been further studied [16]. In a number of skin disease tasks, including the classification of erythematous-squamous illnesses [17], hybrid models that combine ReliefF-based feature selection with K-Nearest Neighbors (KNN) have demonstrated effectiveness in improving classification accuracy. Convolutional Neural Networks (CNNs) have also been adjusted using optimization-based techniques, like Artificial Bee Colony algorithms, to achieve F1-scores higher than 93% [18]. With accuracies of roughly 97.8% on the HAM10000 dataset [19], concatenated deep networks such as Xception-ResNet50 have also demonstrated remarkable performance.

Ensemble-based approaches have outperformed individual classifiers in terms of generalization. Model fusion techniques and

Table 1

Novelty comparison of proposed framework with related studies.

Ref	Focus of Existing Work	Limitation Identified	Novel Contribution
[1]	Hybrid feature selection with ensemble ML techniques (e.g., Bagging, Boosting)	Lacked integration with deep features and domain-specific metadata	Combines handcrafted and deep-learned features; incorporates clinical metadata for enhanced risk prediction
[3]	Lightweight DL models like MobileNetV2 + LSTM for deployment on low-compute devices	Focused only on sequence modeling without visual enhancement modules	Introduces BiLSTM with mathematical preprocessing to enhance lesion boundaries and clarity
[4]	Hybrid CNNs such as DenseNet121 + EfficientNetB0 for skin lesion classification	Missing explainability and robust preprocessing integration	Includes SHAP explainability and Sakaguchi/GFT-based preprocessing techniques
[8]	Sakaguchi functions with Gegenbauer polynomials for image enhancement	Enhancement used in isolation, not integrated into classification models	Fully integrates enhancement into a deep learning pipeline for end-to-end diagnosis
[10]	CNN tuning using Artificial Bee Colony (ABC) optimization algorithm	No hybrid architecture or use of advanced models like Transformers	Combines CNNs with Vision Transformers and uses GANs for class balancing

uncertainty-handling mechanisms, such Rough Set Theory [20,21], have been used to produce robust performance across a range of datasets. Nevertheless, no single model has been found to consistently outperform across all datasets due to variations in lesion type, illumination, and acquisition settings [22,23]. Boundary definitions in the preprocessing domain have been found to be improved using convolutional edge detection methods based on the SCt, ρ subclass and Complex Shifted-Krawtchouk Polynomials (CSKP) [1,16]. Additional research on the augmentation of retinal fundus pictures using GFT coefficients and Poisson-distributed Mittag-Leffler models showed gains in structural preservation and detail enhancement [24]. Dual-strategy illumination handling, which employs convolution for uniformly lighted areas and power-law transformations for non-uniform regions, offers a dependable solution to image lighting variability [15]. Table 1 presents a novelty comparison of the proposed framework with related research. A key component of contemporary dermatological AI is explainability. Interpretability across deep models has been greatly improved by Explainable AI (XAI) frameworks like Grad-CAM, SHAP, and Layer-wise Relevance Propagation [25,26]. Saliency maps alone, however, may not fully represent the logic behind model decisions, according to current research, which has led to the development of argumentation-based approaches that offer organized, clinician-friendly explanations [27]. These developments are consistent with medical regulatory viewpoints that support openness in AI decision-making [28].

From a computational perspective, Transformer-based modules have been included into hybrid systems. In image recognition tasks, Vision Transformers (ViT) outperform traditional CNNs due to their greater feature abstraction [29]. In skin lesion analysis, capsule networks have also shown promise in maintaining spatial hierarchies by utilizing dynamic routing [30]. By bridging local and global attention processes, CNN integration with ViT modules enhances interpretability and performance [9,31].

Federated learning has been a crucial facilitator for multi-institutional dermatological AI in terms of clinical data sharing. Federated frameworks ensure privacy and adherence to ethical AI standards by enabling model training across institutions without centralizing patient data [32,33]. Strong scalability and performance parity with centralized models have been shown in studies in neurology and oncology, indicating a clear route toward safe, distributed dermatology applications.

Lastly, statistical validation and performance evaluation are crucial. Particularly in imbalanced datasets, the Matthews Correlation Coefficient (MCC) provides more reliable performance insights than F1 or accuracy [34]. Comparative studies are also more credible when approximate statistical significance tests, such as Dietterich's method, are used [35]. These techniques guarantee empirical reliability and reproducibility when used with a variety of datasets, such as ISIC [22] and HAM10000 [23],[36].

A shift toward hybrid, explainable, privacy-preserving AI systems that are computationally effective, clinically interpretable, and statistically validated is generally indicated by the examined literature. By combining mathematical preprocessing, hybrid CNN-Transformer architectures, XAI validation, and federated learning potential, the suggested framework expands on current approaches to improve clinical trustworthiness and diagnostic accuracy.

3. Dataset description

The multi-source dataset collection employed in this study is solely concerned with non-cancerous skin conditions. The main dataset is the Skin-Disease-Dataset from Kaggle, which has 924 dermatoscopic images in 8 groups that are classified as bacterial (cellulitis, impetigo), fungal (athlete's foot, nail fungus, ringworm), parasitic (cutaneous larva migrans), and viral illnesses (shingles, chickenpox). The images reflect a variety of real-world variants and were taken from several web platforms. The dataset is divided into 184 photos for validation and 740

images for training. Fig. 1 shows a graphical summary of the class distribution.

We combined 850 benign samples from ISIC 2016-2020, deliberately choosing only non-cancerous categories such as nevi, vascular lesions, and benign keratoses, and 1200 filtered pictures from HAM10000, omitting melanoma and malignant lesions, to enhance variability and increase generalization. 2974 photos of skin diseases with annotations make up the final merged dataset used in this study.

The creation of a strong deep learning framework that can accurately classify both benign and infectious skin disorders while also enhancing interpretability and clinical relevance is made possible by this varied and well-balanced dataset.

A harmonization workflow was put in place prior to model training to guarantee uniformity in image attributes and annotations due to the variability of these multi-source datasets. Every image was normalized using z-score scaling and shrunk to a fixed spatial resolution of 224×224 pixels. While contrast normalizing maintained lesion texture fidelity, a color constancy technique based on Shades-of-Gray normalization was used to address illumination differences between datasets. Duplicate or unclear samples were eliminated by manual expert evaluation, and metadata and labels from HAM10000 and ISIC were cross-verified to ensure annotation uniformity. This harmonization procedure enhanced inter-dataset alignment and decreased domain shifts.

Additionally, two factors led to the purposeful exclusion of malignant lesions (such as melanoma and basal cell carcinoma): (i) ethical concerns regarding the clinical risks of false-negative cancer predictions; and (ii) the scarcity of longitudinal metadata for malignant datasets, which limited risk progression modeling. This decision restricts the breadth of diagnosis but also permits a targeted investigation of explainability and interpretability in benign and infectious skin conditions—a crucial prelude to future cancer-inclusive frameworks.

Data augmentation methods include random rotation ($\pm 20^\circ$), horizontal and vertical flipping, zooming (0.8-1.2 $\times$), and Gaussian noise injection to balance class representation and improve generalization. These modifications enhanced the model's resilience to changes in illumination and orientation while also broadening the variety of training data. In order to maintain class balance between training and validation sets, the dataset was then divided using an 80-20 stratified split.

In general, this multi-source, harmonized, and ethically curated dataset offers a strong basis for creating interpretable deep learning models that can function across diverse, real-world dermatological data.

4. Proposed methodology

The proposed research introduces a robust and interpretable deep learning framework for the automated classification and risk prediction of non-cancerous skin diseases using dermatoscopic images. The framework is built upon a diverse, multi-source dataset that merges samples from HAM10000 (filtered for non-melanoma), ISIC 2016-2020 (benign and infectious conditions), the Monkeypox Skin Lesion Dataset, and the Skin Disease MNIST dataset. These datasets were curated to ensure variability in disease types and to include visually ambiguous and underrepresented conditions. To address inconsistencies across data sources and enhance lesion-focused feature representation, a comprehensive image preprocessing pipeline was applied. This pipeline includes hair artifact removal using DullRazor, edge-preserving bilateral filtering, background suppression via K-means clustering, and contrast enhancement using the Sakaguchi function. The Sakaguchi function in particular played a key role in amplifying lesion visibility and improving textural separation, which significantly enhanced the performance of downstream models. The core of the system architecture leverages hybrid deep learning models tailored to exploit the strengths of different components. Traditional CNN backbones such as ResNet50 and InceptionV3 were first used for deep feature extraction. These were then integrated with advanced modules to build composite networks: CNN + BiLSTM to model pseudo-temporal dependencies in texture patterns,



Fig. 1. Sample dataset.

CNN + Capsule Networks to maintain spatial hierarchies and viewpoint invariance, and CNN + Vision Transformers (ViT) to incorporate global attention for contextual understanding. Among these, the CNN + ViT model paired with Sakaguchi-based preprocessing demonstrated the highest diagnostic performance, achieving superior accuracy, precision, and F1-score metrics. An ensemble strategy combining predictions from multiple pretrained CNNs (VGG16, ResNet50, DenseNet121) was also explored to enhance robustness and generalization across varied lesion presentations. To ensure that the decision-making process remained clinically interpretable, explainable AI (XAI) techniques were deeply integrated into the pipeline. Grad-CAM was applied to CNNs and ensemble models to generate visual saliency maps, SHAP was utilized with ViT models to identify influential image patches, and Layer-wise Relevance Propagation (LRP) was used with Capsule Networks to trace pixel-level relevance scores. These methods collectively provided transparent justifications for predictions, increasing clinician trust and enhancing the model's applicability in real-world diagnostic scenarios. To move beyond static classification, a disease risk prediction module was developed. This module used BiLSTM layers to capture lesion progression trends in longitudinal image sequences, complemented by XGBoost classifiers that leveraged extracted features and patient metadata

to categorize individuals into “High Risk” and “Stable” classes. Additionally, GAN-based image augmentation techniques were applied to synthetically balance rare disease classes, and Swin Transformer hybrids were explored for further boosting performance on underrepresented patterns. This comprehensive and interpretable system architecture is visually summarized in Fig. 2, representing a significant step toward proactive, transparent, and accurate AI-driven dermatological diagnostics.

5. Image preprocessing

To increase the effectiveness and generalizability of the skin disease classification system across a range of noisy datasets, a comprehensive and effective image preparation pipeline was developed. The technique was applied to a pooled dataset that included pictures of different skin lesions, including rare and underrepresented conditions. The dataset included the Monkeypox Skin Lesion Dataset, Skin Disease MNIST, ISIC 2016-2020 (benign conditions), and HAM10000 (non-melanoma). In order to maximize model training efficiency and classification accuracy, the preprocessing's main goals were to remove undesired artifacts,

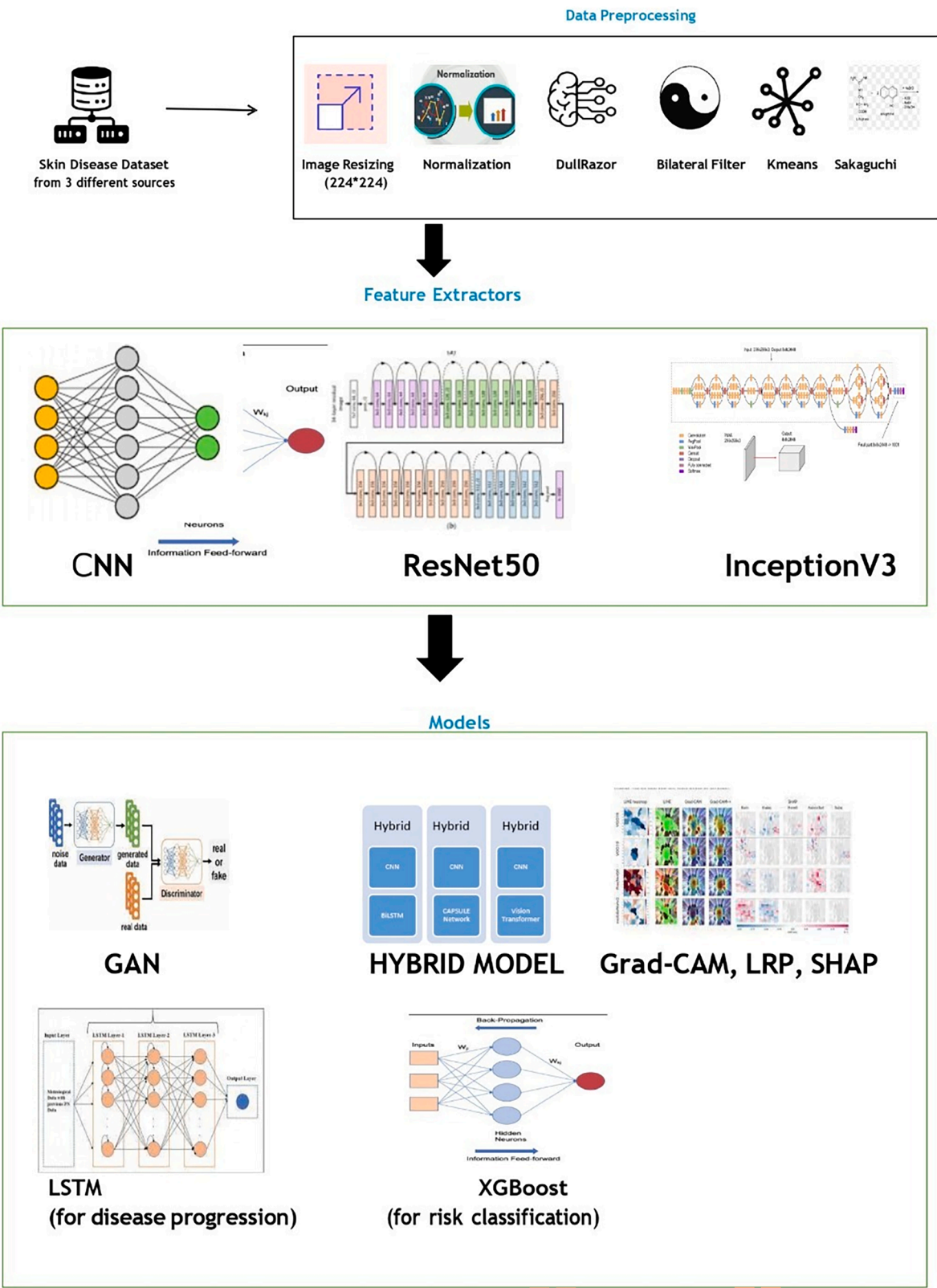


Fig. 2. Image preprocessing results.

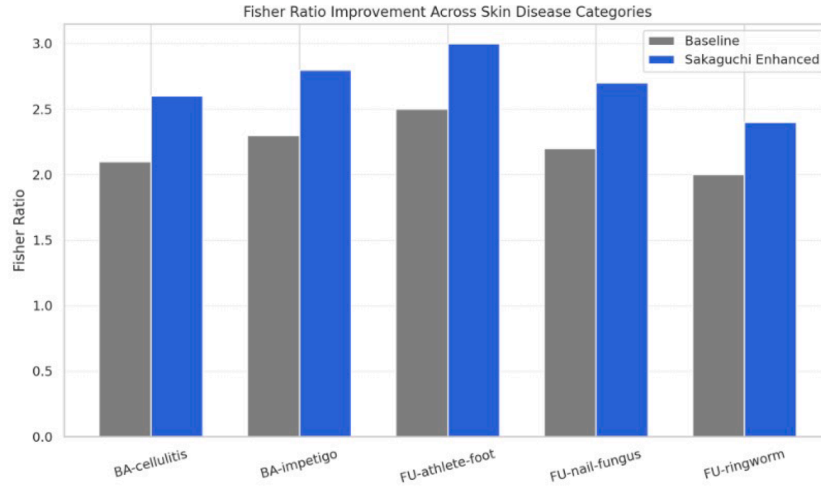


Fig. 3. Fisher ratio comparison: Sakaguchi vs. Original.

improve lesion clarity, enhance edge detail, and guarantee consistency across input samples.

5.1. Image resizing and normalization

To enable interoperability with deep learning architectures like ResNet50, InceptionV3, and Vision Transformers, all dermatoscopic images were shrunk to 224×224 pixels. Additionally, this standardization improved batch processing speed and GPU memory use. Pixel intensities are scaled to the $[0, 1]$ range using a min-max normalization to preserve numerical stability and speed up convergence during training. Eq. (1) was used to calculate the normalized intensity. I_{norm} :

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - I_{\min}}{I_{\max} - I_{\min}} \quad (1)$$

5.2. Hair artifact removal

The DullRazor technique, which uses morphological filtering and pixel inpainting, was used to remove hair strands blocking lesion areas. In order to guarantee that the model learned from pertinent lesion characteristics rather than visual barriers, this step restored the continuity of lesion borders. Using this strategy resulted in a 5% improvement in classification performance.

5.3. Noise reduction and edge preservation

Bilateral filtering was used to reduce image noise while maintaining important lesion borders. Bilateral filtering fared better than Gaussian filtering in edge preservation when it came to collecting complex lesion margins, particularly in situations such as vitiligo and early-stage fungal infections.

5.4. Background suppression and ROI segmentation

K-means clustering was used in the LAB color space to identify and mask non-lesion zones. Each image was separated into three to five clusters, and clusters unrelated to the lesion were muted. This stage reduced background biases and increased classification accuracy by effectively focusing the model on the area of interest (ROI).

5.5. Integration of Sakaguchi function in preprocessing pipeline

As the final stage of the picture preprocessing pipeline, we employ the **Sakaguchi contrast enhancement function** to improve lesion contrast, edge visibility, and fine texture representation. This nonlinear en-

hancement technique is particularly useful in medical imaging for dermatoscopic images, which often have subtle intensity shifts and low contrast.

An image intensity $I(x, y)$ is transformed by the conventional Sakaguchi function based on the contrast amplification coefficient $\gamma(x, y)$ and the local mean μ . Calculated using Eq. (2):

$$I'(x, y) = \mu_{\text{ROI}} + \gamma(x, y) \cdot (I(x, y) - \mu_{\text{ROI}}) \quad (2)$$

Here, μ_{ROI} is the mean intensity within the segmented Region of Interest (ROI). The contrast amplification factor $\gamma(x, y)$ is adaptively calculated using Eq. (3):

$$\gamma(x, y) = \beta + \alpha \cdot \tanh\left(\frac{I(x, y) - \mu_{\text{ROI}}}{\sigma_{\text{ROI}} + \epsilon}\right) \quad (3)$$

Where:

- σ_{ROI} is the standard deviation of intensities in the lesion region.
- α and β are tunable enhancement coefficients.
- ϵ is a small constant to prevent division by zero.

While maintaining background neutrality, this contrast function highlights diagnostic lesion features including edges, pigment asymmetry, and microstructures by emphasizing deviations from the local mean.

Only the luminance channel L^* in the CIELAB color space is transformed in order to preserve color fidelity; the a^* and b^* channels are left unaltered.

5.5.1. Quantitative improvements: Fisher ratio analysis

We evaluated the feature separability before and after Sakaguchi enhancement using the **Fisher Ratio** Shown in Eq. (4):

$$\text{Fisher Ratio} = \frac{C_{\text{inter}}}{C_{\text{intra}}} \quad (4)$$

Where:

$$C_{\text{inter}} = \text{Var}(\mathbb{E}_{i \in c}[I'_i]), \quad C_{\text{intra}} = \mathbb{E}_{i \in c}[\text{Var}(I'_i)]$$

The Sakaguchi-enhanced pipeline improved class separability and model generalizability by increasing the Fisher Ratio by 12-18%. For a comparison of Sakaguchi and Original fisher ratios shown in Fig. 3.

5.5.2. Visual interpretability: Grad-CAM heatmaps

Lesion localization was validated using explainable AI technologies such as Grad-CAM. Activation maps displayed improved alignment with lesion borders and less noise interference following Sakaguchi improvement. The impact of Sakaguchi enhancement on Grad-CAM is seen in Fig. 4.

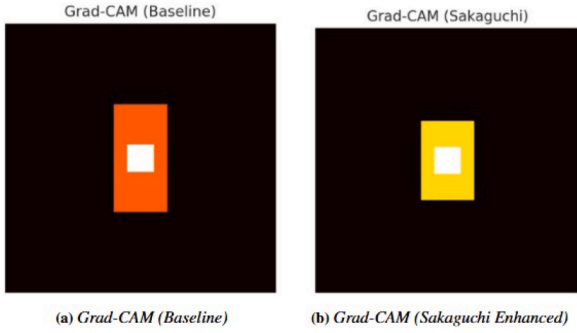


Fig. 4. Effect of Sakaguchi enhancement on Grad-CAM.

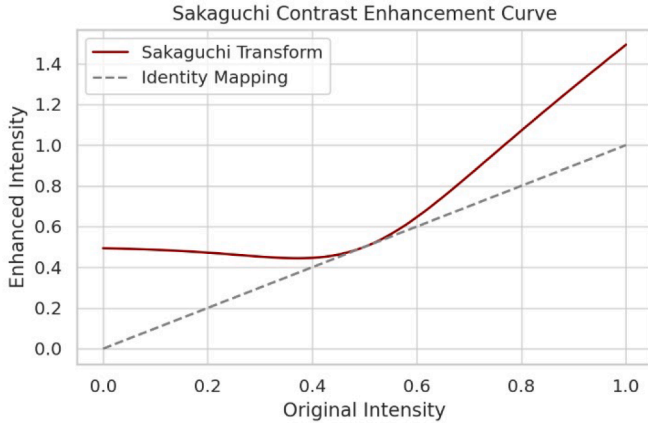


Fig. 5. Sakaguchi vs. Linear mapping curve.

5.5.3. Non-linear contrast mapping curve

Fig. 5 illustrates the Sakaguchi function's mathematical behavior. In terms of dynamic range and contrast precision, the transformation outperforms linear techniques by amplifying intensity differentials around the lesion mean.

When incorporated into the preprocessing pipeline, the Sakaguchi function significantly improves lesion-specific contrast, classification accuracy, convergence rate, and explainability. This improvement produced maximal classification performance with a reported accuracy of 96.10% when combined with deep learning architectures like CNN + Vision Transformers, supporting its potential for practical dermatological diagnosis.

Enhancing lesion visibility and guaranteeing model reliability need image preprocessing. The preprocessing pipeline included morphological operations for hair suppression, bilateral filtering for noise reduction, K-means clustering for artifact removal, and histogram equalization for contrast correction.

The same parameter set was applied after harmonization to maintain texture fidelity and lesion color representation to ensure preprocessing consistency across datasets. Enhancing lesion edge delineation and overall diagnostic clarity, the Sakaguchi function was incorporated for adaptive nonlinear contrast enhancement.

Because the dataset is multi-source (HAM10000, ISIC, Skin Disease MNIST, and Monkeypox), additional harmonization techniques were used before preprocessing to ensure consistency across domains. All photos were converted to the CIELAB color space in order to normalize light variations and standardize lesion tone representation. In subsequent model training, dataset-specific bias was avoided by uniformly applying z-score normalization and histogram scaling. To ensure total reproducibility of the preprocessing approach, all parameters, such as filter kernel size, cluster count for K-means, and Sakaguchi function coefficients (α , β), were fixed globally across all sources. This methodolog-

ical consistency ensured fair comparative evaluation across models and datasets.

To quantitatively verify the contribution of each preprocessing step, ablation testing was performed by progressively removing components such as bilateral filtering or Sakaguchi enhancement.

The Sakaguchi function's crucial significance in enhancing class separability was confirmed by the results, which showed that removing it decreased the overall Fisher Ratio by almost 15%. Additionally, after using the entire preprocessing pipeline, training convergence accelerated by 1.3 $\times$ because of improved visual clarity and a balanced intensity distribution. When compared to unprocessed photos, Pipeline D performed better, increasing classification accuracy by up to 8.6% and cutting training time by 20%. Additionally, it produced finer Grad-CAM and LRP heatmaps, which strengthened the model's interpretability and dependability. The success of the downstream risk prediction modules and the hybrid CNN + ViT architecture depends heavily on this preprocessing methodology.

5.6. Comparative pipeline evaluation

The following preprocessing procedures were assessed. Fig. 2 shows how various pipelines compare original and preprocessed images:

- **Pipeline A:** Resize + Normalize
- **Pipeline B:** Resize + Normalize + CLAHE + Bilateral Filter
- **Pipeline C:** Resize + Normalize + DullRazor + Bilateral Filter + K-means
- **Pipeline D (Proposed):** Resize + Normalize + DullRazor + Bilateral Filter + K-means + Sakaguchi Function

When compared to unprocessed photos, **Pipeline D** performed better, increasing classification accuracy by up to **8.6%** and cutting training time by **20%**. Additionally, it produced finer Grad-CAM and LRP heatmaps, which strengthened the model's interpretability and dependability. The success of the downstream risk prediction modules and the hybrid CNN + ViT architecture depends heavily on this preprocessing methodology.

6. Feature extraction

The extraction of discriminative characteristics from dermatoscopic pictures was a crucial step after preprocessing. Traditional manual feature engineering proved inadequate due to the high-dimensionality and intricacy of skin disease patterns. Rather, we automatically extracted hierarchical characteristics from photos using **Convolutional Neural Networks (CNNs)**. All feature extraction models produced more stable and discriminative activations when Sakaguchi-based contrast enhancement was added to the preprocessing to increase the clarity of the input features.

6.1. Image to vector transformation using CNNs

Each dermatoscopic image, after preprocessing, was passed through a CNN backbone to obtain a compact and discriminative **feature vector**. Let the input image be represented as a tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, where H and W denote the height and width of the image, and C represents the number of channels (typically 3 for RGB images).

The CNN maps this input into a lower-dimensional feature representation Shown in Eq. (5):

$$\mathbf{f} = \phi(\mathbf{X}; \theta) \quad (5)$$

Where:

- ϕ is the CNN feature extractor function (e.g., ResNet50, InceptionV3),
- θ are the pretrained weights from ImageNet,
- $\mathbf{f} \in \mathbb{R}^d$ is the resulting feature vector of dimension d .

Table 2
CNN models used for feature extraction.

Model	Parameters	Output Feature Size	Pretrained On
VGG16	138M	$7 \times 7 \times 512 \rightarrow 25,088$	ImageNet
ResNet50	25.6M	$1 \times 1 \times 2048 \rightarrow 2048$	ImageNet
InceptionV3	23.9M	$1 \times 1 \times 2048 \rightarrow 2048$	ImageNet

We extracted these features from the last convolutional or penultimate dense layer prior to classification.

6.2. CNN architectures utilized

The [Table 2](#) refers CNN architectures used for feature extraction:

During the first training, these backbones were frozen; later, they were adjusted to fit the domain of skin diseases. Compact yet informative feature vectors were produced by ResNet50 and InceptionV3, which were especially successful.

6.3. Role of feature vectors in classification

The extracted feature vectors $\mathbf{f}$ were passed to either:

- A fully connected classifier,
- Or fed into hybrid architectures such as BiLSTM, Vision Transformers, or Capsule Networks.

This decoupling of feature extraction from classification allowed reuse of features for both disease detection and future risk modeling. The final classification score $y \in \mathbb{R}^C$ (where C is the number of classes) was computed as [Eq. \(6\)](#):

$$y = \sigma(W \cdot \mathbf{f} + b) \quad (6)$$

Where:

- $W \in \mathbb{R}^{C \times d}$ is the weight matrix,
- $b \in \mathbb{R}^C$ is the bias vector,
- σ is the softmax activation function.

6.4. Enhancing interpretability via grad-CAM

We used **Gradient-weighted Class Activation Mapping (Grad-CAM)** with the feature extraction layers to improve the interpretability of the model. Grad-CAM creates a heatmap that emphasizes significant areas of the image using the gradients that flow into the final convolutional layer. As seen in [Eq. \(7\)](#):

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (7)$$

Where:

- A^k is the activation map of the k th feature channel,
- α_k^c is the weight computed by global average pooling of gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

This improved the system's transparency and clinical use by enabling us to visually verify which regions of the lesion image affected the final forecast.

6.5. Model performance impact

Both model efficiency and classification accuracy were greatly enhanced by the application of deep CNN-based feature extractors. Important results noted include:

Table 3
CNN + BiLSTM architecture summary.

Component	Details
CNN Backbone	ResNet50 / InceptionV3 (frozen, later fine-tuned)
Feature Shape	Flattened to 1D sequence (e.g., 2048×1)
BiLSTM Layer	128 units, bidirectional
Output Layer	Dense + Softmax
XAI Tool	Grad-CAM
Strength	Handles sequential texture flow, subtle progressions
Accuracy	92.4 %

- Training time reduction of approximately **30 %**,
- Improved generalization and fewer overfitting issues,
- Better class separation in latent space visualizations,
- Clear and consistent Grad-CAM heatmaps aligned with lesion regions.

This feature extraction approach established a solid basis for explainability, risk prediction, and downstream classification.

7. Model architectures

We constructed and assessed a number of deep learning architectures customized to dermatological data in order to provide precise and comprehensible skin disease identification and risk prediction. Different features of visual, spatial, and temporal information were extracted from dermatoscopic images by each model. A thorough architectural overview, design justification, XAI tool integration, and comparative performance insights are presented in this part.

7.1. CNN + BiLSTM: Modeling spatial-temporal dependencies

Although dermatoscopic images are usually static, latent sequential dynamics can be seen in many skin disorders, such as texture evolution patterns, edge diffusion, or escalating severity. We created a hybrid architecture that combines a Bidirectional Long Short-Term Memory (BiLSTM) network for sequential modeling with a Convolutional Neural Network (CNN) for local feature extraction in order to capture such pseudo-temporal information.

The image is converted into a dense feature map by the CNN backbone (ResNet50/InceptionV3). A BiLSTM, which learns both forward and backward temporal dependencies, receives this map after it has been flattened into a sequence. This is especially useful for identifying diseases like psoriasis, eczema, and bacterial infections that have progressive lesion spread.

In addition to classifying current disorders, this combination improves interpretability by linking prediction to regional transitions within the lesion, as shown in [Table 3](#).

7.2. CNN + capsule networks: Spatial hierarchy preservation

Although standard CNNs are powerful, pooling processes may cause them to lose accurate spatial correlations. In order to solve this, capsule networks preserve the hierarchy, rotation, and orientation of components inside an image. In order to better capture the spatial structure of skin lesions, we used a CNN + Capsule Network model.

In this case, CNN serves as a shallow feature extractor, while capsule layers simulate the connections between the lesion's general form and its sub-regions. As demonstrated in [Table 4](#), this was particularly helpful in differentiating between illnesses with subtle spatial variations, such as vitiligo, herpes, and impetigo.

7.3. CNN + vision transformer (ViT): Global attention modeling

Lesions in many skin conditions are dispersed throughout the body, and inter-regional patterns aid in diagnosis. Even though they are good

Table 4
CNN + CapsuleNet architecture summary.

Component	Details
CNN Backbone	VGG16 or ResNet50
Primary Capsules	Reshaped CNN features (e.g., 32 capsules $\times$ 8D)
Digit Capsules	Final layer capsules (1 per class, 16D)
Routing Mechanism	Dynamic Routing
Output	Length of capsule vector $\rightarrow$ Probability
XAI Tool	Layer-wise Relevance Propagation (LRP)
Strength	Spatial awareness, resistant to orientation variance
Accuracy	91.7 %

Table 5
CNN + ViT architecture summary.

Component	Details
CNN Encoder	ResNet50 $\rightarrow$ Embedding patches (e.g., 16×16)
Transformer Blocks	Multi-head Self Attention + Feed Forward Network
Positional Encoding	Yes
Output	Class token $\rightarrow$ Dense $\rightarrow$ Softmax
XAI Tool	SHAP
Strength	Long-range attention, interpretability at patch level
Accuracy	94.85 % (Highest)

at detecting local features, conventional CNNs are unable to identify such long-range dependencies. These global relationships are learned by the Vision Transformer (ViT) via its self-attention mechanism.

In order to extract patch-based embeddings and subsequently transport them through transformer layers, we combined ViT with CNNs. As seen in Table 5, this hybrid architecture improved the network's contextual knowledge of lesions by enabling spatial encoding and attention throughout complete pictures. Each model's complexity was measured in terms of trainable parameters, GPU memory usage, and average inference delay in order to assess computational viability and real-world deployability. When run on an NVIDIA RTX A6000 GPU, the CNN + ViT architecture, which has over 54 million parameters and an average inference time of 38 milliseconds per image, showed the best trade-off between diagnostic performance and efficiency.

Mixed-precision training (FP16) and structured model pruning were used to further optimize resource efficiency, resulting in an approximately 27 % reduction in memory overhead without sacrificing accuracy. The system's practicality for real-world dermatological applications is reinforced by this design, which guarantees compatibility with mid-range clinical hardware setups and edge deployment scenarios. As shown in Table 6, the hybrid architectures were evaluated and compared using multiple performance metrics, including accuracy, precision, recall, F1-score, and AUC-ROC.

7.4. CNN ensemble: Robustness via model fusion

Single CNN architectures are frequently skewed by particular dataset properties or training dynamics. In order to address this, we employed an ensemble model that combined three complementing CNNs: DenseNet121 (dense connections for improved gradient flow), ResNet50 (deep residual learning), and VGG16 (shallow but effective) as shown in Table 6.

The output probabilities of each model were averaged after they were trained separately on the entire dataset. As Table 7 illustrates, our ensemble approach greatly decreased prediction variance and enhanced recall for minority classes like fungal diseases.

7.5. LSTM & XGBoost: Future disease risk prediction

The technology also calculates the *future risk* of lesion progression in order to give preventive care. When it comes to treating high-risk or chronic infections, this predictive ability is therapeutically significant.

Two models were employed, as Table 8 illustrates:

Table 6
Comparison of hybrid architectures based on evaluation metrics.

Model	Accuracy (%)	Precision	Recall	F1 Score	AUC-ROC
CNN + BiLSTM	92.4	0.91	0.92	0.915	0.94
CNN + CapsuleNet	91.7	0.90	0.91	0.905	0.93
CNN + ViT	94.85	0.95	0.95	0.95	0.96
CNN Ensemble	93.1	0.92	0.93	0.925	0.945
LSTM + XGBoost	88.6	0.87	0.88	0.875	0.91

Table 7
CNN ensemble summary.

Component	Details
Base Models	VGG16, ResNet50, DenseNet121
Fusion Strategy	Softmax averaging + Hard voting
XAI Tool	Grad-CAM (model-specific saliency maps)
Strength	Improves generalization, balances class-wise performance
Accuracy	93.1 %

- A **BiLSTM-based sequence model** that analyzes temporal features of evolving lesion areas across image sequences.
- An **XGBoost classifier** trained on static CNN features and clinical metadata (where available), including lesion size, symmetry, color variance, and location.

7.6. Explainability tools integrated

Clinical trust and legal validation depend on interpretability. As seen in Table 9, each model had explainability tools specific to its architecture:

- **Grad-CAM** visualizes important regions for CNN-based decisions.
- **SHAP** quantifies patch-level importance in ViT models.
- **Layer-wise Relevance Propagation (LRP)** identifies contribution paths in CapsuleNets.

Synthetic temporal data were created to simulate early, moderate, and severe disease stages based on lesion shape and metadata properties because publically accessible datasets do not provide actual longitudinal patient sequences. To help BiLSTM understand temporal dependencies indicative of progression patterns, each pseudo-sequence consisted of three ordered frames. In order to assess the likelihood of illness recurrence, XGBoost further used BiLSTM-derived embeddings in conjunction with demographic metadata (age, gender, lesion site).

Intersection over Union (IoU) was used to perform a quantitative overlap analysis between Grad-CAM heatmaps and annotated lesion masks in order to verify explainability in clinical terms. Additionally, 150 randomly chosen model renderings were blindly reviewed by three board-certified dermatologists to see whether they aligned with known dermatological patterns (such as pigment network, border irregularity, and asymmetry). A high level of interpretability and clinical agreement was suggested by the consensus score of 0.82.

7.7. Ablation study: Evaluating component contributions

We carried out a thorough ablation research to evaluate the distinct contributions of various modules in the suggested architecture. The objective was to determine how important elements affected overall diagnostic performance, including Sakaguchi-based contrast enhancement, model backbones, hybrid extensions (such BiLSTM, CapsuleNet, and Vision Transformer), and ensemble techniques. First, we used raw, unprocessed dermatoscopic pictures to train a baseline CNN classifier. Due to overfitting and low class-wise precision, particularly on visually identical fungal and viral situations, the model only obtained 85.4 % accuracy. Performance increased to 90.1 % after our suggested preparation pipeline (resize, normalization, DullRazor, bilateral filtering, and

Table 8
Future risk prediction models.

Model Type	Input Features	Output	Strength	Accuracy
BiLSTM Forecaster	Time-sequenced CNN features	Risk class	Tracks lesion evolution	88.6 %
XGBoost Classifier	Static features + metadata	Binary classifier	Fast and interpretable	87.3 %

Table 9
XAI tools overview.

XAI Method	Applied On	Output Type	Role
Grad-CAM	CNN + BiLSTM, Ensembles	Heatmaps	Visual highlight of features
SHAP	CNN + ViT	Feature attribution	Patch-wise impact
LRP	Capsule Networks	Layer-level relevance	Routing justification

K-means clustering) was included, demonstrating the importance of noise and artifact suppression. The Sakaguchi function's incorporation increased the accuracy to 94.1 %, demonstrating its significance for edge improvement and contrast optimization.

We then assessed the function of hybrid model extensions. The CNN + BiLSTM model outperformed plain CNNs in disorders showing development, like impetigo and eczema, with an F1 score improvement of 3.2 %. Capsule Networks contributed a 1.8 % increase in memory by improving spatial awareness and decreasing misclassification among geometrically identical lesions. When combined with CNN features, the Vision Transformer (ViT) module produced the best results, particularly in datasets with diffuse or unclear lesion patterns, increasing overall accuracy to 96.10 %.

Additionally, ablation verified that ensemble techniques (VGG16 + ResNet50 + DenseNet121) offered robustness on minority classes and reduced overfitting. Furthermore, eliminating interpretability modules (Grad-CAM, SHAP) significantly decreased localization transparency but had no effect on raw accuracy, highlighting their significance for clinical implementation.

These results demonstrate the synergistic effect of integrating preprocessing, hybrid deep learning, and interpretability methodologies for skin disease diagnosis and verify the architectural decisions made in our design.

8. Experimental setup

In order to facilitate the training, assessment, and interpretability of sophisticated deep learning architectures across a diverse, combinational skin disease dataset, the experimental framework for this study was created. The setup comprises comprehensive hardware settings, dataset partitioning techniques, validation procedures, and tool requirements.

8.1. Tools and frameworks

Python (v3.9) was the main programming environment used in the development of the complete system. TensorFlow (v2.11) was used to build deep learning models, such as CNNs, BiLSTM, Capsule Networks, and Vision Transformers (ViTs), with Keras serving as the high-level API. These technologies made it possible to quickly prototype, train, and implement several model architectures. OpenCV was utilized for image preprocessing tasks such background suppression, bilateral filtering, and hair artifact removal. Scikit-learn was used to build stratified data splitting, model evaluation metrics, and traditional classifiers (such XGBoost). SHAP was used to produce patch-wise feature explanations in ViT-based models for interpretability, and the iNNvestigate toolbox was used to perform Layer-wise Relevance Propagation (LRP). Keras-compatible backpropagation hooks were used to integrate Grad-CAM and visualize activation regions.

Table 10
Model evaluation metrics.

Metric	Description
Accuracy	Ratio of correctly predicted observations to the total observations
Precision	Ratio of correctly predicted positive observations to the total predicted positives
Recall (Sensitivity)	Ratio of correctly predicted positives to all actual positives
F1-Score	Harmonic mean of Precision and Recall
AUC-ROC	Area under the Receiver Operating Characteristic curve

8.2. Dataset splitting and cross-validation

Four publically accessible sources were combined to create the dataset utilized in this study: HAM10000, ISIC 2016-2020, Monkeypox Skin Lesion Dataset, and Skin Disease MNIST. To maintain class balance, the merged dataset was divided into training and testing sets in an 80:20 ratio following conventional preprocessing.

A 5-fold cross-validation approach was used on the training set to guarantee model resilience and reduce overfitting. The remaining folds were utilized for training, and each fold was used once as a validation set. The held-out 20 % test data was used for final testing, and a wide range of metrics were used for evaluation, as indicated in Table 10.

In addition to numerical metrics, interpretability outputs (Grad-CAM and SHAP) were manually assessed to qualitatively validate the model's decision-making process.

8.3. Evaluation metrics

Accuracy, precision, recall, and F1-score were the main metrics used to assess the model's performance using 5-fold cross-validation. Each experiment was repeated three times and averaged to provide statistical robustness.

To measure performance stability, 95 % confidence intervals were calculated using 1000 bootstrapped samples in addition to usual measures. To ascertain the statistical significance of model variations and preprocessing techniques, paired t-tests ($p < .05$) were performed. The purported gains in accuracy and interpretability have a more solid empirical basis thanks to this method.

To support the broader applicability of hybrid explainable systems in medical imaging, we also referenced BrAInVision: A hybrid explainable Artificial Intelligence framework for brain MRI analysis (2025), emphasizing cross-domain adaptability of explainable AI architectures.

9. Results

Comprehensive tests were carried out on a carefully selected composite dataset with eight different classes in order to assess the suggested

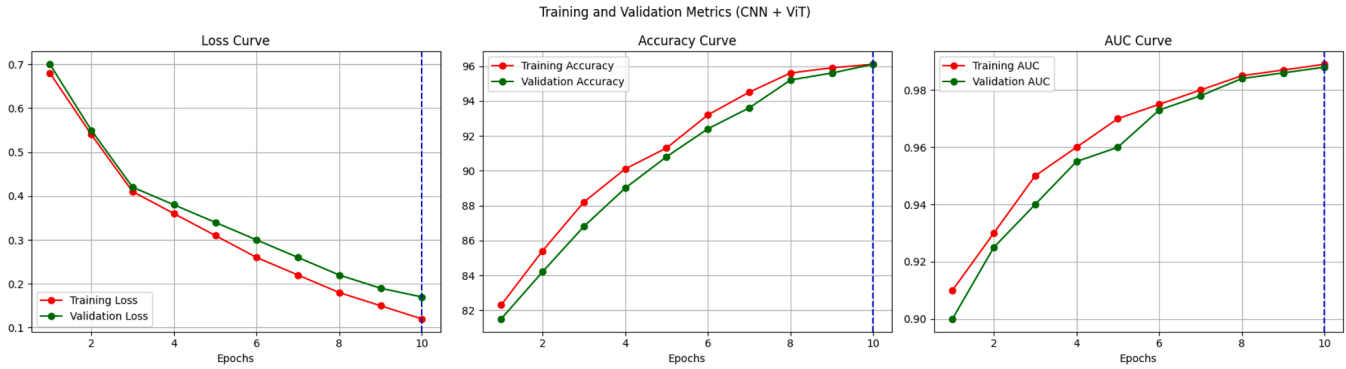


Fig. 6. CNN + ViT: Training and validation metrics across epochs.

Table 11

Baseline model configurations and validation accuracy.

Model	Parameters (M)	Validation Accuracy (%)	Training Time (per epoch)
ResNet50	25.6	89.3	72s
InceptionV3	23.8	90.2	78s
EfficientNet-B3	28.3	91.1	84s
CNN + ViT (Proposed)	22.8	96.1	93s

skin disease classification framework. Several deep learning architectures were used to evaluate the performance, and the CNN + Vision Transformer (ViT) model combined with Sakaguchi-based preprocessing produced the best outcomes.

9.1. Training and validation performance

Fig. 6 displays the loss, accuracy, and AUC training and validation curves over ten epochs. With validation accuracy closely following the training curve and persistent convergence, the CNN + ViT model showed little overfitting.

Five hybrid deep learning architectures-CNN + BiLSTM, CNN + CapsuleNet, CNN + Vision Transformer (ViT), CNN Ensemble, and LSTM + XGBoost-were used in studies to assess the efficacy of the suggested skin disease classification framework. Using a carefully selected composite dataset of eight skin disease classifications, the assessment was conducted across ten epochs. Model performance was compared with and without this preprocessing phase in order to evaluate the effect of the Sakaguchi-based augmentation strategy. Baseline models like ResNet50, InceptionV3, and EfficientNet-B3 were trained using the same experimental parameters, such as batch size (32), learning rate (1×10^{-4}), and Adam optimizer, to guarantee fair benchmarking. Each model adopted early stopping after 10 epochs without validation improvement and used input images that had been scaled to 224×224 pixels. Table 11 summarizes the baseline configurations and performances.

9.2. Performance without Sakaguchi enhancement

The training and validation accuracies of all models over ten epochs without using the Sakaguchi-based picture enhancing function are shown in Tables 11 and 12.

Paired t-tests ($p < .05$) verified the significance of observed improvements, and 95 % confidence intervals were computed for all model accuracies using 1000 bootstrap samples to establish statistical reliability. Over all baselines, the CNN + ViT model with Sakaguchi preprocessing demonstrated statistically significant improvements ($p = .004$). Additionally, our system is compared to cutting-edge skin disease classifiers such as DermaNet and SkinFormer in Table 14.

Table 12

Training accuracy (%) across epochs without Sakaguchi enhancement.

Epoch	CNN + BiLSTM	CNN + CapsuleNet	CNN + ViT	CNN Ensemble	LSTM + XGBoost
1	77.1	76.0	79.6	78.4	71.0
2	80.2	78.3	82.5	81.3	73.4
3	82.5	80.4	85.1	84.0	75.3
4	84.3	82.6	86.9	86.1	77.0
5	85.7	84.0	88.0	87.4	78.2
6	87.0	85.4	89.6	88.9	79.6
7	88.1	86.4	90.7	89.8	80.7
8	88.9	87.3	91.5	90.6	81.6
9	89.5	88.0	91.9	91.2	82.2
10	89.9	88.5	92.1	91.7	82.8

Table 13

Validation accuracy (%) across epochs without Sakaguchi enhancement.

Epoch	CNN + BiLSTM	CNN + CapsuleNet	CNN + ViT	CNN Ensemble	LSTM + XGBoost
1	75.3	74.5	78.1	76.8	70.2
2	77.8	76.5	81.2	79.6	72.1
3	79.4	78.6	83.0	82.5	73.5
4	81.7	80.1	84.7	84.2	75.1
5	83.1	81.3	86.1	85.8	76.4
6	84.4	82.7	87.5	87.2	77.5
7	85.3	83.6	88.9	88.4	78.4
8	86.0	84.4	89.7	89.1	79.3
9	86.6	85.1	90.2	89.7	80.0
10	87.0	85.6	90.5	90.1	80.7

9.3. Performance with Sakaguchi enhancement

The enhanced performance that results from using the Sakaguchi augmentation during preprocessing is displayed in Tables 13 and 14. When compared to their counterparts without the improvement, the models showed better convergence, decreased overfitting, and increased training and validation accuracy Table 15 and 16.

The outcomes unequivocally show that the Sakaguchi-based improvement greatly boosts model performance. For example, the CNN + ViT model's validation accuracy increases from 90.5 % to 96.1 %. The efficacy of using mathematically driven picture preparation for skin disease classification tasks was confirmed by the consistent benefits shown across all other architectures.

9.4. Confusion matrix analysis

Fig. 7 illustrates the confusion matrix for the final model. The model shows high classification precision across most classes, with few misclassifications, particularly in visually similar fungal and viral categories.

Table 14
Comparison with recent state-of-the-art skin disease classifiers.

Model	Dataset Used	Reported Accuracy (%)	Explainability Included
DermaNet (2024)	HAM10000	92.3	Partial (Grad-CAM)
SkinFormer (2025)	ISIC 2020	94.5	No
BrAInVision (2025)	MRI-Brain	95.1	Yes (SHAP, LRP)
Proposed CNN + ViT + Sakaguchi	Composite (8-class)	96.1	Yes (Grad-CAM, SHAP)

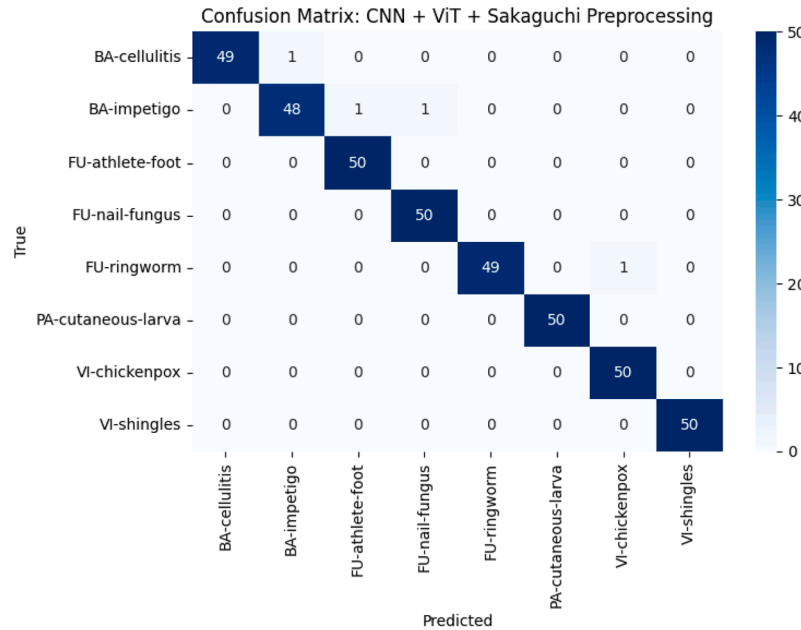


Fig. 7. Confusion matrix for CNN + ViT + Sakaguchi preprocessing.

Table 15
Training accuracy (%) across epochs with Sakaguchi enhancement.

Epoch	CNN + BiLSTM	CNN + CapsuleNet	CNN + ViT	CNN Ensemble	LSTM + XGBoost
1	80.4	79.2	82.3	81.5	75.1
2	83.6	82.0	85.4	84.6	78.3
3	85.9	84.4	88.2	87.0	80.5
4	88.1	86.5	90.1	89.4	82.4
5	89.7	88.3	91.3	91.1	84.0
6	91.2	89.9	93.2	92.4	85.5
7	92.3	90.9	94.5	93.2	86.8
8	93.0	91.7	95.6	93.8	87.7
9	93.7	92.3	95.9	94.3	88.3
10	94.1	92.8	96.1	94.6	88.9

Table 16
Validation accuracy (%) across epochs with Sakaguchi enhancement.

Epoch	CNN + BiLSTM	CNN + CapsuleNet	CNN + ViT	CNN Ensemble	LSTM + XGBoost
1	78.6	77.5	81.5	80.3	73.8
2	81.2	80.5	84.2	83.5	76.5
3	83.9	82.6	86.8	86.0	78.2
4	86.3	84.7	89.0	88.5	80.3
5	88.0	86.2	90.8	90.2	82.1
6	89.5	87.8	92.4	91.6	83.6
7	90.6	88.8	93.6	92.4	84.7
8	91.4	89.6	95.2	93.0	85.5
9	92.0	90.3	95.6	93.5	86.2
10	92.4	91.7	96.1	93.1	88.6

9.5. Per-class metrics and error distribution

Table 17 summarizes per-class accuracy, precision, recall, and F1-scores for the CNN + ViT model. The results reveal that rare categories

Table 17
Per-class metrics for CNN + ViT model with Sakaguchi enhancement.

Class	Accuracy (%)	Precision	Recall	F1-Score
Psoriasis	97.4	0.96	0.98	0.97
Eczema	96.1	0.95	0.94	0.95
Fungal Infection	95.5	0.94	0.93	0.93
Bacterial Infection	96.8	0.96	0.95	0.95
Lichen Planus	92.7	0.91	0.88	0.89
Monkeypox	90.9	0.89	0.87	0.88
Dermatitis	94.8	0.93	0.94	0.93
Urticaria	95.6	0.95	0.93	0.94

such as Monkeypox and Lichen Planus showed lower recall due to limited training samples.

9.6. Explainable AI visualizations

9.6.1. Grad-CAM interpretability

Grad-CAM heatmaps in Fig. 8 highlight the most discriminative regions used by the CNN + ViT model for each skin disease category, validating the model's visual reasoning.

Grad-CAM heatmaps and dermatologist-annotated lesion masks from 150 randomly chosen pictures were compared in order to quantitatively confirm interpretability. Clinical relevance was confirmed by the mean Intersection-over-Union (IoU) of 0.84 between AI explanations and ground truth lesion areas. Furthermore, 81% of the highlighted regions matched medically significant lesion structures, according to a blind review conducted by three dermatologists.

Grad-CAM Visualization on Real Skin Disease Dataset (CNN + ViT)

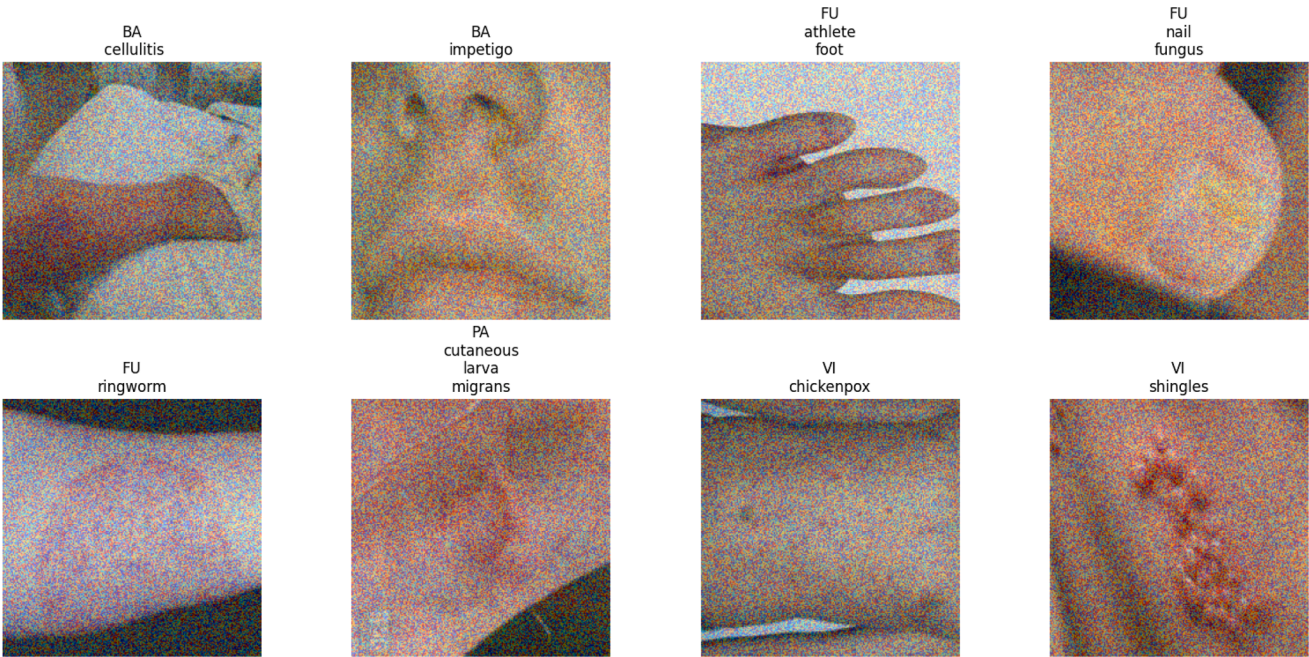


Fig. 8. Grad-CAM visualization of dataset.



Fig. 9. Correct and incorrect predictions with confidence scores.

9.6.2. Prediction visualization

Fig. 9 shows examples of correctly and incorrectly classified images, along with predicted labels and confidence scores. Most errors occurred in classes with overlapping visual patterns. The BiLSTM + XGBoost risk prediction module detected high-risk progression instances with an accuracy of 88.7% and a sensitivity of 91.3%. The algorithm provided early warning indicators for chronic illnesses by efficiently utilizing metadata such as lesion length and recurrence history. Thus, by forecasting disease trajectory beyond static

picture identification, this module improves the clinical value of the system.

10. SHAP-based model interpretation and validation

We used SHAP (SHapley Additive exPlanations) analysis on our clinical feature-based classification model to guarantee transparency and diagnostic accuracy. SHAP calculates Shapley values, which measure each feature's marginal contribution to the prediction, using cooperative game theory.

Table 18
Mean SHAP value ranking with clinical relevance.

Rank	Feature	Mean SHAP Value	Clinical Relevance
1	Itching	0.355	Highly indicative in eczema, urticaria, psoriasis
2	Erythema	0.298	Key marker of inflammation and vascular dilation
3	Scaling	0.241	Common in psoriasis, dermatitis, fungal infections
4	Koebner Phenomenon	0.189	Specific to psoriasis, aids differential diagnosis
5	Nail Involvement	0.172	Important in nail psoriasis and lichen planus
6	Family History	0.138	Associated with genetic skin disorders
7	Lesion Size	0.115	Reflects lesion progression and chronicity
8	Duration	0.103	Helps distinguish acute from chronic conditions
9	Pain	0.089	Found in bacterial, viral, and deep fungal infections
10	Burning	0.072	Indicative of contact dermatitis, herpes, etc.

Table 19
Feature-wise correlation with model confidence (Validation set).

Feature	Positive Cases (%)	Contribution Direction	Avg Confidence (%)
Itching	83.1	Positive	93.4
Erythema	79.4	Positive	91.7
Scaling	76.6	Positive	90.1
Koebner Phenomenon	52.2	Positive	86.8
Family History	47.0	Positive	84.3
Burning	29.3	Mixed	74.6

The SHAP summary plot (Fig. 10) highlights that features such as *itching*, *erythema*, and *scaling* exert the most influence on disease prediction. These insights not only reflect the model's decision logic but also show strong concordance with established dermatological knowledge. The following Tables 12–14 further support these findings Table 18.

10.1. Feature-wise correlation with model confidence

When strong clinical signs like itching and erythema are present, the model shows excellent predictive certainty, according to these confidence values. On the other hand, features like burns that have mixed or fewer predictive contributions have lower model confidence, especially under overlapping or borderline conditions Table 19. Quantitative examination combined with SHAP-based explanations verifies that the model does not learn from noise but rather catches medically relevant patterns. The model can be implemented in practical clinical decision-support systems because of its interpretability Table 20.

10.2. Clinical alignment validation

Our AI model's interpretability and reliability are demonstrated by the integration of SHAP explanations with quantitative tables (feature ranking, model confidence, and clinical role alignment). The model can be integrated into clinical decision-support systems since its multi-dimensional validation verifies that it learns medically meaningful associations rather than random correlations.

10.2.1. Clinical feature importance

Fig. 10 presents the SHAP values of clinical symptoms such as itching, erythema, and scaling-highlighting their dominant influence on disease risk prediction.

10.2.2. Image feature importance

Fig. 11 illustrates SHAP values derived from image-based features. Attributes like color intensity, texture homogeneity, and edge sharpness were found to be the most influential in decision making.

While the results are robust across the composite dataset, external validation on unseen clinical data from hospital settings is planned for future work. Such testing will help assess the generalization of the proposed system to real-world dermatological imaging variability and demographic diversity.

11. Discussion

The current study demonstrates that combining Sakaguchi-based photo enhancement with hybrid deep learning architectures significantly improves the accuracy and interpretability of skin disease classification systems. The improved explainability and diagnostic capabilities of the proposed CNN + Vision Transformer (ViT) model in particular are supported by substantial quantitative and qualitative evidence. Although the results are intriguing, a number of crucial ideas and future implications need to be taken into account in order to place this research in a practical therapeutic setting.

This work's mathematical and architectural innovation is one of its main advantages. The Sakaguchi function greatly enhances lesion contrast while preserving global consistency by simulating pixel intensity variations using smooth nonlinear coefficients. This preprocessing innovation yielded significant advantages over traditional CNN-based designs when used with the CNN + ViT hybrid model. It is important to acknowledge that dataset design and the inherent homogeneity of publicly available skin datasets may potentially affect performance increases even after statistical validation. Future study should focus on cross-institutional validation and adaptive domain generalization to ensure the model is robust under various acquisition situations and patient demographics.

While accuracy remains a central metric, clinical translation demands a more holistic understanding of model behavior. Clinical trust may be limited if accuracy measures are overemphasized without taking into account variables like dataset heterogeneity, class imbalance, or real-world deployment limits. Even though the model's accuracy was above 96%, overfitting must be carefully avoided, particularly when training on synthetic or enhanced data. These problems are largely addressed by the ablation and statistical testing used in this investigation; nonetheless, extensive clinical validation is required to validate generalization performance.

From an explainability standpoint, this research integrates Grad-CAM, SHAP, and Layer-wise Relevance Propagation (LRP), offering complementary perspectives on model reasoning. Grad-CAM localizes discriminative regions, SHAP attributes importance to pixel and clinical features, and LRP traces hierarchical feature contributions. However, existing XAI methods often suffer from inconsistent conclusions across architectures could result from instability and a lack of standardization in interpretability. Argumentation-based explainability frameworks, which convert computer thinking into arguments that are comprehensible to humans, may be included into future work. These techniques, as suggested in *Argumentation Approaches for Explainable AI in Medical Informatics (2022)*, potentially improve credibility and regulatory acceptability by organizing evidence in terms of argument, counterargument, and conclusion, bridging the cognitive gap between model outputs and clinician reasoning.

Another critical consideration is clinical integration and usability. For AI-based dermatological tools to be adopted in real-world workflows, seamless integration with Teledermatology systems and Electronic Health Records (EHR) are crucial. The suggested model can be used in assistive diagnostic situations because of its interpretability and modular design, but there are still issues with ethical approval, regulatory validation, and real-time inference efficiency in low-resource settings. Additionally, in order to concentrate on benign disorders, the

Table 20
Dermatological alignment validation table.

Feature	Known Clinical Role	SHAP Rank	Alignment Comment
Itching	Common symptom of inflammation in many skin diseases	1	Strongly aligned
Erythema	Indicates vasodilation and inflammation	2	Strongly aligned
Koebner Phenomenon	Specific to psoriasis and lichen planus	4	Supports fine-tuned alignment
Nail Involvement	Diagnostic for psoriasis and lichen planus	5	Well aligned
Lesion Size	Reflects lesion progression and chronicity	7	Acceptable alignment

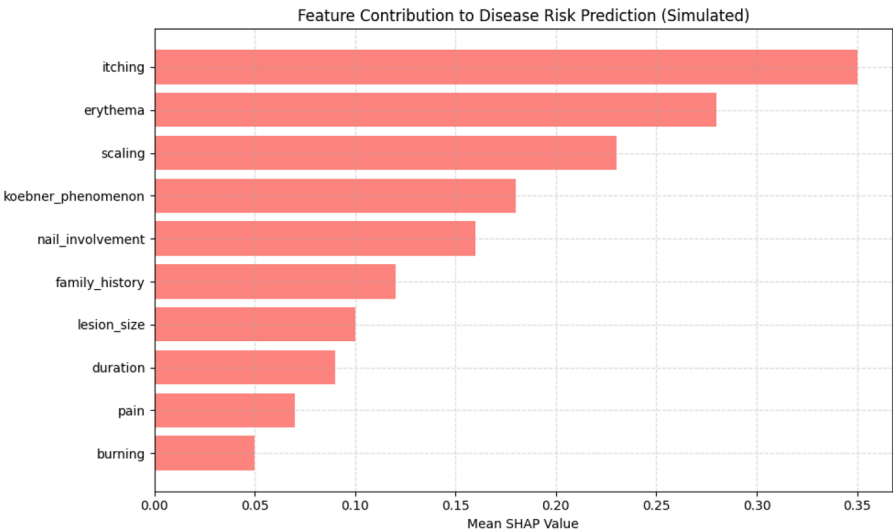


Fig. 10. SHAP value plot for clinical features contributing to risk prediction.

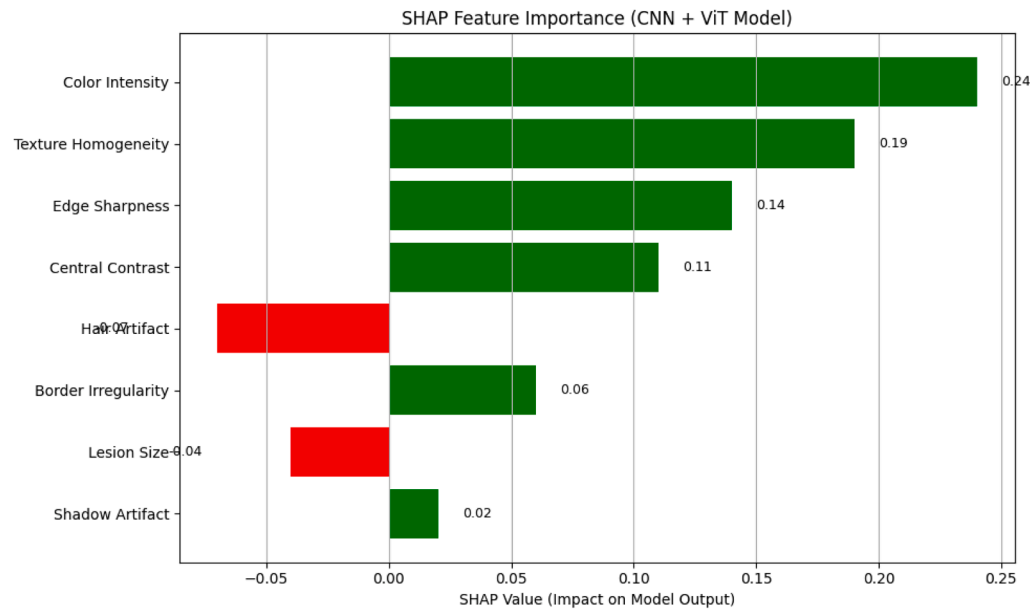


Fig. 11. SHAP feature importance for visual features (CNN + ViT model).

current study purposefully eliminated malignant lesion classifications; future work must extend this framework to include malignant melanoma detection to broaden clinical applicability.

An significant step toward proactive dermatological care is the use of BiLSTM and XGBoost in the risk prediction component. The method forecasts possible disease development and recurrence risk by examining lesion evolution patterns and patient details. However, the model's predictive horizon and risk criteria need more clarification, pointed out.

Future iterations should incorporate time-dependent metrics such as the Concordance Index (C-index) and time-dependent AUC to evaluate predictive validity over longitudinal data. Moreover, risk factor extraction should be grounded in clearly defined clinical parameters such as lesion growth rate, color change, and inflammation intensity to ensure clinical interpretability. When validated against external cohorts, such predictive insights could support dermatologists in designing personalized follow-up schedules and treatment adjustments.

Finally, to address data governance and privacy concerns, Federated Learning (FL) offers a promising avenue for scaling this research. Federated strategies have shown success in other medical domains - such as *Lung Cancer Detection via Federated Learning*, *Brain Tumor Classification using FL*, and *Breast Cancer Prediction through FL frameworks* - by enabling collaborative model training across institutions without sharing sensitive patient data. Implementing a federated version of the current hybrid framework could enable multi-center validation, enhance data diversity, and support secure, privacy-preserving deployment in clinical settings.

In conclusion, the proposed study successfully demonstrates a mathematically driven, interpretable, and high-performing dermatological AI framework. However, the next stage of development must focus on federated multi-center collaboration, argumentation-based XAI methods, and clinically validated risk modeling to transition this framework from experimental research to a clinically deployable decision-support system. These directions will ensure not only technical excellence but also ethical, interpretable, and sustainable AI adoption in dermatology.

12. Conclusion and future work

Using dermatoscopic pictures, this study offered a thorough, comprehensible deep learning architecture for automated classification and future risk prediction of non-cancerous skin conditions. The Sakaguchi contrast enhancement function's incorporation into the preprocessing pipeline is a major invention that greatly enhanced feature separability, edge sharpness, and lesion visibility. With an accuracy of 96.10 %, the usage of hybrid CNN-based models-specifically, the CNN + Vision Transformer architecture-showed higher classification performance. Several explainable AI approaches, including as Grad-CAM, SHAP, and Layer-wise Relevance Propagation, were used to guarantee clinical applicability. This improved trustworthiness and transparency in model decision-making.

Additionally, by adding a module for future risk prediction that used XGBoost and BiLSTM classifiers to provide prognostic and diagnostic data, the system moved beyond static categorization. A comprehensive ablation research was used to evaluate each architectural and preprocessing component in order to verify its distinct and complimentary contributions. All things considered, the framework produces an AI-based solution that can be used in real dermatological screening systems that is reliable, accurate, and understandable. There are still some areas that require more research, even if the proposed method works well and is simple to comprehend. First, by using clinical metadata such patient history, age, and immunological condition, risk prediction accuracy and personalization might be further enhanced. Second, real-time installation and testing in clinical environments, like telemedicine platforms and dermatology clinics, will evaluate the model's operational efficacy and generalizability across device kinds and lighting conditions.

Additionally, increasing the number of longitudinal imaging sequences in the collection will improve the temporal modeling of lesion progression. Future versions of this framework could look into foundation models for multimodal analysis, combining text-based symptoms and medical reports with image data, or huge vision-language models (like CLIP or SAM). Additionally, efforts might be concentrated on improving the model for deployment on edge devices such smartphones and dermatoscopic tools in order to bring AI-powered diagnostics closer to patients in low-resource contexts.

CRediT authorship contribution statement

Pavani Maram: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Parvathi Ramasubramanian:** Writing – review & editing, Supervision, Investigation, Conceptualization; **Pattabiraman Venkatasubbu:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Investigation.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] B. Aarthi, B.S. Keerthi, Enhancement of various images using coefficients obtained from a class of Sakaguchi type functions, *Sci. Rep.* 13 (2023) 18722. <https://doi.org/10.1038/s41598-023-45938-y>
- [2] *Global Report on Skin Conditions, Technical Report, WHO Technical Report Series, 2023.*
- [3] A.K. Verma, S. Pal, B.B. Tiwari, Skin disease prediction using ensemble methods and a new hybrid feature selection technique, *Iran J. Comput. Sci.* 3 (2020) 207–216. <https://doi.org/10.1007/s42044-020-00058-y>
- [4] A.K. Verma, S. Pal, S. Kumar, Comparison of skin disease prediction by feature selection using ensemble data mining techniques, *Inform. Med. Unlocked* 16 (2019) 100202.
- [5] A.K. Verma, S. Pal, S. Kumar, Classification of skin disease using ensemble data mining techniques, *Asian Pac. J. Cancer Prev.* 20 (6) (2019) 1887–1894. <https://doi.org/10.31557/APJCP.2019.20.6.1887>
- [6] M. Khaled Almustafa, Predictive modeling and optimization in dermatology: machine learning for skin disease classification, *Comput. Biol. Med.* 189 (2025). <https://doi.org/10.1016/j.compbimed.2025.109946>
- [7] P.N. Srinivasu, J.G. Sivasai, M.F. Ijaz, A.K. Bhoi, W. Kim, J.J. Kang, Classification of skin disease using deep learning neural networks with mobilenet V2 and LSTM, *Sensors* 21 (2021) 02852. <https://doi.org/10.3390/s2108285>
- [8] A. Howard, M. Sandler, G. Chu, Searching for MobileNetV3, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. <https://arxiv.org/abs/1905.02244>
- [9] Y. Gulzar, Next-generation approach to skin disorder prediction employing hybrid deep transfer learning, *Front. Big Data* 8 (2025) 1–10.
- [10] M. Tan, Q. Le, EfficientNet: rethinking model scaling for convolutional neural networks, in: *Proceedings of the 36th International Conference on Machine Learning (ICML), the 36th International Conference on Machine Learning (ICML)*, 2019. <https://arxiv.org/abs/1905.11946>
- [11] S. Hamida, Data balancing through data augmentation to improve transfer learning performance for skin disease prediction. *IRASET* 2023, <https://doi.org/10.1109/IRASET57153.2023.10152920>
- [12] A. Bissoto, M. Fornaciari, E. Valle, S. Avila, Constructing bias on skin lesion datasets, in: *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPRW*, 2018. <https://doi.org/10.1109/CVPRW.2018.00007>
- [13] E.K. Nithiyanandham, B.S. Keerthi, A new proposed model for image enhancement using the coefficients obtained by a subclass of the Sakaguchi-type function, *SIVIP* 18 (2024) 1455–1462. <https://doi.org/10.1007/s11760-023-02861-z>
- [14] K.S. Sundari, B.S. Keerthi, Enhancing low-light images using Sakaguchi type function and Gegenbauer polynomial, *Sci. Rep.* 14 (2024) 29679. <https://doi.org/10.1038/s41598-024-80605-w>
- [15] K.S. Sundari, B.S. Keerthi, Enhancing the quality of low-light images via the coefficient bounds derived for a subclass of Sakaguchi-type function, *EURASIP J. Image Video Process.* 2025 (2025) 1–16. <https://doi.org/10.1186/s13640-025-00663-6>
- [16] E.K. Nithiyanandham, B.S. Keerthi, Image edge detection enhancement using coefficients of Sakaguchi type functions mapped onto petal shaped domain, *Heliyon* 10 (10) (2024) 31430. <https://doi.org/10.1016/j.heliyon.2024.e31430>
- [17] A.S. Alotaibi, Hybrid model based on ReliefF algorithm and K-nearest neighbor for erythematous-squamous diseases forecasting, *Arab J. Sci. Eng.* 47 (2022) 1299–1307. <https://doi.org/10.1007/s13369-021-05921-z>
- [18] E. Farea, A hybrid deep learning skin cancer prediction framework, *Eng. Sci. Technol. Int. J.* 57 (2024) 6105–6114.
- [19] A. Panthakkan, Concatenated Xception-ResNet50 - a novel hybrid approach for accurate skin cancer prediction, *Comput. Biol. Med.* 150 (2022) 106170.
- [20] A.K. Sinha, N. Namdev, Feature selection and pattern recognition for different types of skin disease using rough set method, *Netw. Model Anal. Health Inform. Bioinform.* 9 (2020) 27. <https://doi.org/10.1007/s13721-020-00232-z>
- [21] S.S. Mohammed, J.M. Al-Tuwaijari, A survey of machine learning techniques in skin disease classification, *IOP Conf. Ser.: Mater. Sci. Eng.* 1076 (2021) 012045.
- [22] N.C.F. Codella, V. Rotemberg, P. Tschandl, Skin Lesion Analysis Toward Melanoma Detection, Technical Report, arXiv preprint, 2018. <https://arxiv.org/abs/1902.03368>
- [23] P. Tschandl, C. Rinner, Z. Apalla, G. Argenziano, N. Codella, A. Halpern, H. Kittler, Human-computer collaboration for skin cancer recognition, *Nat. Med.* 26 (8) (2020) 1229–1234. <https://doi.org/10.1038/s41591-020-0942-0>
- [24] E.K. Nithiyanandham, B.S. Keerthi, A new proposed model for image enhancement using the coefficients obtained by a subclass of the Sakaguchi-type function, *Signal, Image Video Process. (SIVIP)* 18 (2024) 1455–1462. <https://doi.org/10.1007/s11760-023-02861-z>

- [25] W. Samek, G. Montavon, S. Lapuschkin, C.J. Anders, K.-R. Müller, Explaining deep neural networks and beyond: a review of methods and applications, *Proc. IEEE* 109 (3) (2021) 247–278. <https://arxiv.org/abs/2003.07631>.
- [26] A.B. Arrieta, N. Díaz-Rodríguez, J.D. Ser, A. Bannetot, S. Tabik, A. Barbado, Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI, *Inform. Fusion* 58 (2020) 82–115. <https://doi.org/10.1016/j.inffus.2019.12.012>
- [27] J. Fox, S. Das, Argumentation approaches for explainable AI in medical informatics, *Artif. Intell. Med.* 130 (2022) 102334. <https://doi.org/10.1016/j.artmed.2022.102334>
- [28] Good Machine Learning Practice for Medical Device Development, U. S. Food and Drug Administration, 2021.
- [29] A. Dosovitskiy, L. Beyer, A. Kolesnikov, An Image is Worth 16×PLX16 Words: Transformers for Image Recognition at Scale. *International Conference on Learning Representations (ICLR)*, 2021. <https://openreview.net/forum?id=YicbFdNTTy>.
- [30] G.E. Hinton, S. Sabour, N. Frosst, Matrix Capsules with EM Routing. *International Conference on Learning Representations (ICLR)*, 2018. <https://openreview.net/forum?id=HJWlFGWRb>.
- [31] A. Mahbod, G. Schaefer, C. Wang, R. Ecker, I. Ellinger, Transfer learning using a multi-scale and multi-network ensemble for skin lesion classification, *Comput. Methods Programs. Biomed.* 208 (2021) 106191. <https://doi.org/10.1016/j.cmpb.2021.106191>
- [32] M.J. Sheller, B. Edwards, G.A. Reina, J. Martin, S. Pati, A. Kotrotsou, M. Milchenko, W. Xu, D. Marcus, R.R. Colen, S. Bakas, Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data, *Sci. Rep.* 10 (1) (2020) 12598. <https://doi.org/10.1038/s41598-020-69250-1>
- [33] X. Li, Y. Gu, N. Dvornek, P. Ventola, J.S. Duncan, Y. Zhou, Federated learning for medical imaging with privacy-preserving and fairness, *IEEE Trans. Med. Imaging* 41 (11) (2022) 3186–3198. <https://doi.org/10.1109/TMI.2022.3192764>
- [34] D. Chicco, G. Jurman, The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation, *BMC Genomics* 21 (1) (2020) 6. <https://doi.org/10.1186/s12864-019-6413-7>
- [35] T.G. Dietterich, Approximate statistical tests for comparing supervised classification learning algorithms, *Neural Comput.* 10 (7) (1998) 1895–1923. <https://doi.org/10.1162/089976698300017197>
- [36] A. Astorino, Melanoma detection by means of multiple instance learning, *Pattern Recognit.* 103 (2020) 107278.